



PROJECT REPORT

Intelligent Road Pothole Detection & Severity Analysis Platform

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1. Introduction

Road potholes are among the most pervasive causes of vehicle damage, road accidents, and increased commute times worldwide. According to infrastructure maintenance studies, reactive pothole-repair cycles — relying on citizen complaints or periodic manual surveys — are both cost-inefficient and temporally delayed, often leaving dangerous surface anomalies unaddressed for weeks or months.

The emergence of lightweight, real-time capable object detection architectures — particularly the YOLO (You Only Look Once) family — has created an opportunity to automate road inspection at scale. A dashcam or roadside camera unit running an optimised YOLO model can continuously scan the road surface and flag defects without human intervention, feeding detections directly into maintenance management systems.

This work presents PaveMind AI, a production-ready intelligent road pothole detection and severity analysis platform. The key research contributions of this paper are:

- A fine-tuned YOLOv12 Nano (2.6 M parameter) model trained on a merged 765-image dataset achieving mAP@0.5 of 91.23%, outperforming the base model by a significant margin.
- A camera-geometry-based physical size estimation engine that computes real-world pothole dimensions in centimetres without any 3D sensor hardware.
- A multi-factor severity scoring algorithm combining physical size, depth proxy, detection confidence, aspect ratio, and temporal persistence into a single weighted score.
- A seven-stage false-positive mitigation pipeline that suppresses common non-pothole detections including sky, clouds, tyre tracks, and pixel noise.
- A complete end-to-end system integrating a RESTful FastAPI backend, MongoDB Atlas cloud database with local offline fallback, and an interactive React 19 analytics dashboard with GPS heatmapping.

2. Related Work

Early approaches to automated pothole detection relied on vibration-based sensors (accelerometers mounted in vehicles) to detect surface anomalies from road-induced jolts. While effective in controlled conditions, such methods are highly sensitive to vehicle suspension characteristics and speed, producing inconsistent results across different vehicle types.

Image-based detection methods gained traction with the rise of convolutional neural networks. ResNet and VGG-based classifiers were applied to sliding-window detection but suffered from slow inference speeds incompatible with real-time requirements. The introduction of the YOLO architecture by Redmon et al. (2016) unified detection and classification into a single forward pass, enabling real-time performance. Subsequent versions (YOLOv5 through YOLOv12) progressively improved accuracy and efficiency through architectural innovations.

The Road Damage Detection Challenge (CRDDC/RDD2022) benchmark by Arya et al. (2022) provided a large-scale, multi-national annotated dataset covering multiple damage types including potholes (class D40), cracks, rutting, and patches. It established standardised evaluation protocols that this work adopts. Recent literature indicates that YOLOv12, with its Area Attention (A^2) module and Reversed ELAN architecture, offers a superior accuracy-efficiency tradeoff over its predecessors for road-surface anomaly detection tasks.

However, existing published systems typically address only bounding box localisation and do not provide real-world physical dimension estimates, multi-tier severity classification, or end-to-end deployment pipelines. PaveMind AI addresses all of these gaps.

3. Dataset Preparation

3.1 Data Sources

To maximise model generalisation across diverse road conditions, two publicly available annotated datasets were merged:

Source	Images	Annotations	Format	Notes
Kaggle Pothole Dataset	665	~1,350	YOLO (class 0)	Varied lighting & road types
CRDDC / RDD2022	100 (D40 subset)	~674	YOLO (class 3 → 0)	Multi-country; India, USA, Japan
Merged Total	765	2,024	YOLO (class 0)	Single-class: pothole

3.2 Preprocessing & Split

RDD2022 annotation files were remapped — retaining only class D40 (pothole) instances and reassigning them to class index 0 for consistency with the baseline dataset. File-name prefixing prevented collision during merging. The combined dataset was then automatically partitioned into a 70 / 20 / 10 train–validation–test split.

Split	Images	Pothole Instances	Avg / Image
Train	535	1,386	2.59
Validation	153	438	2.86
Test	77	200	2.60
Total	765	2,024	2.64

All images were resized to 640×640 pixels prior to training. Standard YOLO augmentation strategies — mosaic patching, horizontal flipping, HSV colour jitter, and random scaling — were applied during training to improve resilience across varied environmental conditions.

4. Model Architecture & Training

4.1 YOLOv12 Nano

The YOLOv12n (Nano) variant was selected as the detection backbone based on its superior accuracy-to-parameter ratio. Key architectural innovations over previous YOLO generations include:

- Area Attention (A^2) Module — a novel spatial attention mechanism that partitions the feature map into horizontal and vertical regions, enabling efficient long-range dependency modelling without the quadratic cost of standard self-attention.
- Reversed ELAN (R-ELAN) — a reversed residual pathway for feature reuse that improves gradient flow and training stability during fine-tuning.
- Edge-Optimised Design — the Nano variant targets deployment on consumer-grade CPU and edge hardware while maintaining competitive accuracy.

Architectural Property	Value
Model Variant	YOLOv12n (Nano)
Fused Layers	159
Total Parameters	2,556,923 (~2.6 M)
Computation Load	6.3 GFLOPs
Input Resolution	640 × 640 px
Output Classes	1 (pothole)

4.2 Training Configuration

Transfer learning from COCO pre-trained weights (yolov12n.pt) was employed. Training ran for up to 25 epochs with an early-stopping patience of 5 epochs. A batch size of 4 was used to accommodate CPU-only training environments, with automatic CUDA utilisation when available. After training convergence, the model was additionally exported to ONNX format for optional edge inference deployment. The primary production weights reside at runs/detect/yolov12n_pothole_detector5/weights/best.pt.

5. System Design & Methodology

5.1 False-Positive Mitigation Pipeline

A critical challenge in road-surface detection is the model's tendency to flag visually similar non-pothole objects. A seven-stage post-processing validation pipeline was designed and implemented to reject spurious detections prior to any downstream analysis:

Gate	Rejection Criterion	Target False-Positive Class
Confidence	Score < 0.45	Shadows, image noise
Horizon	Lower edge y < 40% of frame height	Clouds, sky regions
Sky	Upper edge y < 30% of frame height	Background hills
Aspect Ratio	Ratio < 0.80 or > 5.00	Tyre tracks, narrow cracks
Scale	Area < 150 px ²	Pixel-level noise, gravel
Coverage	Box area > 25% of frame	Windshield, bonnet obstruction

5.2 Physical Dimension Estimation

Real-world pothole dimensions are estimated using standard perspective projection geometry, requiring no 3D depth sensor. Given calibrated camera parameters (mounting height, focal length, tilt angle), the system computes the distance to each detection and derives its physical width and road-plane depth in centimetres. Default calibration assumes a 1.3 m camera height, 800 px focal length, and 18° tilt angle — all configurable at runtime. This transforms a pure pixel-space bounding box into an actionable engineering measurement.

5.3 Multi-Factor Severity Scoring

Each validated detection is classified into one of three severity tiers — MINOR, MODERATE, or SEVERE — via a weighted scoring model combining five independent signals:

Scoring Factor	Weight	Description
Physical Size	0.45	Normalised estimated width (primary driver)
Depth Proxy	0.20	Normalised road-plane depth estimate
Detection Confidence	0.15	YOLO output confidence score
Aspect Ratio	0.10	Deviation from ideal pothole circularity
Temporal Persistence	0.10	Consistency across consecutive video frames

Scores below 0.45 are classified MINOR; scores 0.45–0.70 as MODERATE; and scores 0.70 and above as SEVERE. Severity is surfaced to the driver via a dynamic road-condition alert banner rendered on each processed frame.

5.4 Road Health Index

A scan-level Road Health Index (RHI, 0–100) is computed as: $RHI = \max(0, 100 - 15 \times N_{\text{severe}} - 6 \times N_{\text{moderate}} - 2 \times N_{\text{minor}})$, where N represents the count of detections at each severity level. This provides a single interpretable metric suitable for maintenance prioritisation dashboards and automated alert thresholds.

6. End-to-End System Architecture

PaveMind AI is structured across three integrated layers:

6.1 Core Analysis Engine

The ML inference and analysis core (Python) encapsulates four independently importable modules: the SeverityEngine (false-positive filtering, physical estimation, severity scoring), the AlertSystem (dynamic road condition banner generation), the AnalyticsManager (per-run report export in JSON, CSV, and TXT formats), and the Visualizer (OpenCV-based frame overlay rendering with coloured bounding boxes, label chips, and real-time FPS display).

6.2 REST API Backend

A FastAPI application exposes eight REST endpoints for image upload, video upload, scan management, analytics retrieval, and ZIP report download. Video processing runs as a non-blocking background task using ByteTrack multi-object tracking for consistent frame-to-frame pothole identity assignment. Output video is H.264 encoded (yuv420p) for native browser playback compatibility. The database layer uses MongoDB Atlas as the primary store with transparent fallback to a local JSON file store when cloud connectivity is unavailable.

6.3 Web Dashboard

The frontend is a React 19 + Vite single-page application featuring a premium dark-themed UI. Key dashboard sections include: a KPI overview with summary statistics and road health distribution charts; an Image Analyzer supporting multi-file drag-and-drop upload with side-by-side source and annotated frame comparison; a Video Tracker with real-time processing progress and H.264 video playback; and a historical Scans Database with

search, filter, and per-scan ZIP report download. An interactive Leaflet.js GPS heatmap displays all tagged detections as severity-coloured markers.

7. Experimental Results & Evaluation

7.1 Model Benchmark

Two training configurations were benchmarked under identical evaluation conditions using the held-out test split:

Metric	YOLOv12n — 25 Epochs (Selected)	YOLOv12n — 2 Epochs (Baseline)	Improvement
Precision	0.8804 (88.04%)	0.6443 (64.43%)	+36.6%
Recall	0.8105 (81.05%)	0.5342 (53.42%)	+51.8%
mAP@0.5	0.9123 (91.23%)	0.5827 (58.27%)	+56.6%
mAP@0.5:0.95	0.6402 (64.02%)	0.3017 (30.17%)	+112.2%
Avg CPU Latency	281.43 ms	279.53 ms	—
Throughput (CPU)	3.55 FPS	3.58 FPS	—

The 25-epoch model was selected as the production deployment configuration, offering a 56.6% improvement in mAP@0.5 with no meaningful throughput penalty. The 2-epoch model requires substantially more training epochs to reach comparable accuracy on the merged dataset.

7.2 Live Detection Performance

System evaluation on a real Indian road video sequence (692 frames) yielded 2,302 total detections with an average inference latency of 181.9 ms per frame (5.5 FPS on CPU). Severity breakdown across the full sequence: 852 SEVERE (37.0%), 731 MINOR (31.8%), 719 MODERATE (31.2%). Mean detection confidence was 0.57 and average estimated physical pothole width was 73.7 cm. A sample detection on an Indian road image identified 17 simultaneous potholes across all three severity tiers with individual width estimates ranging from 30 cm to 375 cm.

7.3 False-Positive Suppression Efficacy

All seven identified false-positive categories — including clouds, background hills, tyre tracks, road seams, pixel noise, windshield occlusion, and low-confidence shadows — were fully eliminated by the post-processing gate pipeline. An additional logic bug in the AlertSystem where severity counts defaulting to zero caused a permanent SEVERE banner on clean road frames was identified and resolved, improving user-facing alert accuracy to correctly reflect actual detection state.

8. Conclusion & Future Scope

8.1 Conclusion

PaveMind AI demonstrates that a lightweight, low-parameter deep learning model (YOLOv12 Nano at 2.6 M parameters) — when fine-tuned on a well-curated merged dataset and augmented with a principled post-processing and analysis pipeline — can achieve production-grade pothole detection accuracy (mAP@0.5: 91.23%, Precision: 88.04%) suitable for real-world road inspection deployment. The camera-geometry-based

physical size estimator provides actionable engineering measurements without 3D sensor hardware. The end-to-end cloud-integrated platform makes the system immediately deployable in municipal road maintenance contexts.

8.2 Future Scope

Enhancement	Description
GPU Inference	Deploy on CUDA-capable server for 30+ FPS real-time video analysis
Monocular Depth	Integrate MiDaS depth estimation network to replace geometric approximation
Mobile Application	React Native companion app for field inspection teams
Road Authority API	Integration endpoints for municipal road management systems and automated SMS/email alerts
Cloud Deployment	Host frontend on Vercel and backend on Hugging Face Spaces for public access

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